

Lesson 4

Array & For Loop



Store many values, then loop through them.

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- 01 Quick review of Lesson 3
 - 02 Array: create, write, read & clear
 - 03 For loop
 - 04 Array + for loop: average & largest value
 - 05 Classwork: handling data
-

The mini calculator in three steps.

- 1 How to ask users to input two numbers?
How to store them?

```
Serial.println("Please type your first number.");  
while(Serial.available()==0){}  
int a = Serial.parseInt();
```

- 2 How to add the two numbers? Store it
to where?

```
int sum = a+b;
```

- 3 How to show the answer?

```
Serial.print("The answer is:");  
Serial.println(sum);
```

Sensor readings need cleaning.

FUNCTIONS FOR SENSORS

```
incipe.getTemperature()  
incipe.getHumidity()  
incipe.getDistance()  
incipe.getLightIntensity()  
incipe.getPPM()  
incipe.getCOPPM()  
incipe.getAlcoholPPM()  
incipe.getFlameIntensity()
```

DATA CLEANING

Fixing or removing incorrect or incomplete raw data. Here, we focus on interpreting the raw data to make it more useful.

```
if AQvalue < xxxx  
    print Good  
else if AQvalue > xxxx and < xxxx  
    print Moderate  
else  
    print Poor
```

Which sensor does not require data cleaning?

THINK ABOUT WHAT EACH GIVES BACK

`incipe.getLightIntensity()` → 700, 1000...

`incipe.getPPM()` → a ppm number

`incipe.getButtonResponse()` → 1 or 0

ANSWER

Button

WHY

It already gives 1 (pressed) or 0 (not pressed), so there is nothing to interpret.

A collection of variables, like a database.

22	24	23
[0]	[1]	[2]

One name holds many values. Each box has an **index**, and we count the index from **0** instead of 1.

```
int data[3] = {22, 24, 23};
```

Create an array: type, name, size.

```
int data[6];
```

int array with a size of 6, without existing data

```
int value[5] = {2, 4, -8, 3, 2};
```

int array with a size of 5, with existing data

```
float number[2] = {2.8943, 10.23189};
```

You can use other data types too: float, size of 2

Write, read & clear by index.

Write

```
int data[6] = {1,2,3,4,5,6};  
data[1] = 10;
```

→ {1, **10**, 3, 4, 5, 6}

data[1] is the second element. To change the first, use [0].

Read

```
int data[6] = {1,10,3,4,5,6};  
x = data[4];
```

→ x = **5**

x gets a copy of the fifth element. The array does not change.

Clear

```
int data[6] = {1,10,3,4,5,6};  
data[5] = 0;
```

→ {1, 10, 3, 4, 5, **0**}

We assume 0 means nothing, so this removes the sixth element.

What would be the new array?

READ THE CODE

```
int numbers[5] = {1,2,3,4,5};  
numbers[2] = numbers[1] + numbers[4];
```

ANSWER

{1,2,7,4,5}

WHY

numbers[1] is 2 and numbers[4] is 5. Their sum, 7, replaces numbers[2], the third element.

What will be printed?

READ THE CODE

```
float data[8];  
int x = 0;  
data[2] = 1.23;  
data[5] = 2.34;  
data[7] = 3.45;  
x = data[1] + data[5];  
Serial.println(x);
```

ANSWER

2

WHY

data[1] was never set, so it is 0. $0 + 2.34 = 2.34$, and x is an int, so the decimal part is dropped.

Repeat a command with a fixed structure.

A for loop repeats some command. It is mainly used to control arrays.

```
for (int i = 0; i ≤ 25; i++)  
{  
    Serial.println(i);  
}
```

int i = 0

Initial value of a variable to control the loop

i ≤ 25

Used to determine the end of the loop

i++

Change of the value after each loop (i++ adds 1)

What will be printed?

READ THE CODE

```
for (int i = 1; i < 10; i = i+2){  
    Serial.print(i);  
}
```

ANSWER

13579

WHY

i starts at 1 and goes up by 2: 1, 3, 5, 7, 9. At 11 the loop stops. print keeps them on one line.

Use i as the index.

```
int data[5] = {0,1,2,3,4};

void setup() {
  Serial.begin(9600);
}

void loop() {
  for (int i = 0; i < 5; i = i+1)
  {
    Serial.println(data[i]);
    delay(1000);
  }
}
```

i = 0

Initial value of i is 0, the first index

i < 5

Limit i to below 5, the size of the array

data[i]

Prints data[0] the 1st time, data[1] the 2nd time, and so on

What is the value of a?

READ THE CODE

```
int a = 0;
int database[3] = {1,2,3};
for (int i = 1; i < 3; i++){
    database[0] = database[0] + database[i];
}
a = database[0];
```

ANSWER

6

WHY

i = 1: $1 + 2 = 3$. i = 2: $3 + 3 = 6$. database[0] collects the total, so a is 6.

Average & largest value of 100 readings.

AVERAGE

```
for (int i = 1; i < 100; i++){  
    database[0] = database[0] + database[i];  
    database[i] = 0;  
}  
float mean = database[0]/100;  
Serial.println(mean);
```

Add every element into database[0], clearing each one, then divide by 100.

LARGEST VALUE

```
for (int i = 1; i < 100; i++){  
    if (database[0] < database[i]){  
        database[0] = database[i];  
    }  
}  
Serial.print(database[0]);
```

Whenever a bigger value turns up, keep it in database[0].

Handling data.

YOUR NOTES

Refer to your note page 8.

YOU WILL NEED

An array to store the data, and a for loop to go through it.

Three things to take away.

- 1 An array holds many values under one name; the index counts from 0.
- 2 A for loop has a start, an end condition and a step: `for (int i = 0; i < 5; i++)`.
- 3 Together, `data[i]` inside a for loop reads every element: to print, total or find the largest.

Thank you!